

# STAT 579: Course Schedule

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This page provides a detailed week-by-week schedule for STAT 579. Readings should be completed *before* the first class of the week.

- **HR:** Hernán, M. A., & Robins, J. M. (2024). *Causal Inference: What If*.
- **Neal:** Neal, B. (2020). *Introduction to Causal Inference*.
- **Cunningham:** Cunningham, S. (2021). *Causal Inference: The Mixtape*.
- **VanderWeele:** VanderWeele, T. J. (2015). *Explanation in Causal Inference*.

## Part I: Foundations of Causal Inference

| Week | Dates      | Core Topic                     | Primary Readings                            | Optional Reading (Cunningham)  | Project Connection                 |
|------|------------|--------------------------------|---------------------------------------------|--------------------------------|------------------------------------|
| 1    | Aug 19, 21 | What is a Causal Effect?       | <b>HR:</b> Ch. 1–2;<br><b>Neal:</b> Ch. 1–2 | Ch. 2: Potential Outcomes      | Define your causal question.       |
| 2    | Aug 26, 28 | DAGs & Assumption Transparency | <b>Neal:</b> Ch. 3;<br><b>HR:</b> Ch. 6     | Ch. 3: Directed Acyclic Graphs | Draw a preliminary DAG.            |
| 3    | Sep 2, 4   | Exchangeability, Positivity    | <b>HR:</b> Ch. 7;<br><b>Neal:</b> Ch. 4     | Ch. 4: Regression              | Identify assumptions for your DAG. |
| 4    | Sep 9, 11  | Bias (Confounding, Selection)  | <b>HR:</b> Ch. 8–9                          | -                              | Consider biases in your dataset.   |

## Part II: Estimation Methods

| Week | Dates      | Core Topic        | Primary Readings                               | Optional Reading (Cunningham)       | Project Connection          |
|------|------------|-------------------|------------------------------------------------|-------------------------------------|-----------------------------|
| 5    | Sep 16, 18 | Propensity Scores | <b>HR:</b> Ch. 12–13;<br><b>Neal:</b> Ch. 7    | Ch. 5: Matching & Subclassification | Select adjustment strategy. |
| 6    | Sep 23, 25 | IPTW & MSMs       | <b>HR:</b> Ch. 15;<br><b>Neal:</b> Ch. 7.5–7.6 | -                                   | Evaluate MSM relevance.     |

| Week | Dates         | Core Topic             | Primary Readings                         | Optional Reading (Cunningham)    | Project Connection          |
|------|---------------|------------------------|------------------------------------------|----------------------------------|-----------------------------|
| 7    | Sep 30, Oct 2 | G-computation          | <b>HR:</b> Ch. 20–21                     | -                                | Explore g-methods.          |
| 8    | Oct 7         | Instrumental Variables | <b>HR:</b> Ch. 16;<br><b>Neal:</b> Ch. 9 | Ch. 8:<br>Instrumental Variables | Assess natural experiments. |

*Note: UNM Fall Break is Oct 9–10. Thursday's class is cancelled.*

### Part III: Advanced Topics & Applications

| Week | Dates      | Core Topic                 | Primary Readings                                 | Project Connection                      |
|------|------------|----------------------------|--------------------------------------------------|-----------------------------------------|
| 9    | Oct 14, 16 | Effect Modification        | <b>HR:</b> Ch. 4–5; <b>Neal:</b> Ch. 10          | <b>Milestone: Project Proposal Due.</b> |
| 10   | Oct 21, 23 | Mediation I (Foundations)  | <b>VanderWeele:</b> Ch. 1–2; <b>Neal:</b> Ch. 14 | Refine DAG with mediators.              |
| 11   | Oct 28, 30 | Mediation II (Application) | <b>VanderWeele:</b> Ch. 3; <b>HR:</b> Ch. 23     | -                                       |
| 12   | Nov 4, 6   | Sensitivity Analysis       | <b>Neal:</b> Ch. 8; <b>HR:</b> Ch. 14            | Plan robustness checks.                 |
| 13   | Nov 11, 13 | Target Trial Emulation     | <b>HR:</b> Ch. 22                                | Reframe project as a trial.             |
| 14   | Nov 18, 20 | <b>Final Presentations</b> | -                                                | Present your findings.                  |

*Note: Thanksgiving Break is Nov 27–28.*

**Finals Week (Dec 8–13):** Final Project Paper Due.